

QR Tag Scan Flow

Lost Pet Alert System — Reunification via Safety Tag

PawGuard Platform | pawguard-backend (FastAPI + PostgreSQL + Redis + ARQ) | Prepared for Team Review

1. Purpose

A pet wearing a PawGuard safety tag can be identified the moment it's found — before a “found pet” report even needs to be filed — by simply scanning the printed QR code with any phone camera.

2. End-to-End Flow



1. The QR image encodes a public URL containing the raw token (e.g. `pawguard.app/tags/scan?t=RAW_TOKEN`). No app install or login is required.
2. The landing page calls `POST /companion-pets/safety-tag/scan` with the raw token.
3. Backend hashes the token (SHA-256) and looks up `pet_safety_tags.token_hash` — an exact, indexed match.
4. If found and `is_active = true`, the backend returns the pet's emergency-safe profile: name, species, breed, photo, and an owner-written emergency note.
5. `scan_count` is incremented and `last_scanned_at` is stamped, so the owner sees “last scanned 4 minutes ago” in their app — real-time reassurance that someone has the pet.
6. The scan page offers a one-tap “I found this pet” action that pre-fills a Found Report (`POST /lost-found/found`), feeding straight into the matching engine described in the Lost Pet Alert System doc.

3. What the Finder Never Sees

- Owner's name, phone number, or home address — deliberately excluded from the scan-response schema.
- Any authentication token, account data, or ability to edit the pet's record.

Contact between finder and owner is brokered by the platform (in-app messaging / claim workflow), not by exposing raw PII on a public page.

4. Abuse Protection

Control	Value
Endpoint visibility	Public, unauthenticated (by design)
Rate limit	20 requests / 60 seconds per IP (Redis-backed)
Token exposure	Only the SHA-256 hash is ever stored server-side
Revocation	Owner can rotate the tag (old token instantly invalid) if the collar is lost/stolen

5. Sequence Summary

Step	Actor	Action
1	Finder	Scans QR on pet collar
2	Client	Opens public scan page → calls scan API
3	Backend	Hash-match, rate-limit check, returns safe profile
4	Finder	Optionally files a Found Report
5	Matching engine	Scores it against open Lost Reports

Source: `companion_pet/router.py (scan_safety_tag)`, `companion_pet/service.py`, `lost_found/service.py`.